

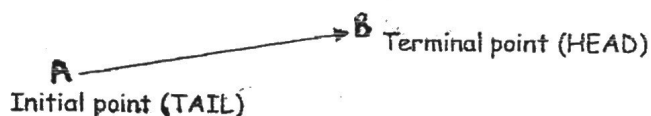
VECTOR CONCEPTS

Vectors are quantities having both magnitude and direction. For example, a car is traveling with a velocity of 30 km/h from the north-west.

Scalars are quantities having magnitude only. Temperature, mass and height are examples as there is no direction associated with these.

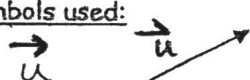
Note: Displacement is a vector quantity (because it refers to traveling a certain distance in a particular direction), whereas distance is a scalar quantity (because there is no direction associated with it).

Representing a vector:



Length and direction should be reflected in the vector diagram. Thus, a vector with twice the magnitude of another vector should be drawn with two times the length.

Symbols used:



bold font

note: some text books/websites use $\mathbf{u}$ & $\mathbf{v}$

Note: For length (magnitude), we use absolute value brackets when we have positive scalars.

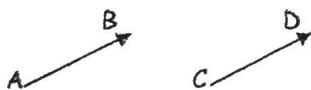
For example, $|\mathbf{u}|$ & $|\mathbf{v}|$ are the magnitudes of

$\mathbf{u}$ & $\mathbf{v}$

Vector Equality:

iff

Two vectors are equal if and only if they have the same magnitude and same direction.

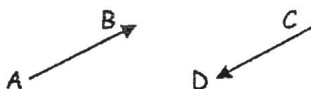


$\vec{AB} = \vec{CD}$ ← direction/magnitude

$|\vec{AB}| = |\vec{CD}|$ ← magnitude

$|\vec{AB}| = |\vec{DC}|$ ← magnitude

Two vectors are opposite if they have the same magnitude but point in opposite directions



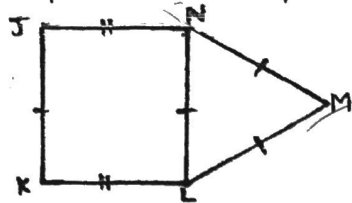
$\vec{AB} = -\vec{CD}$ ← direction

$|\vec{AB}| = |\vec{CD}|$ ← magnitude

When two vectors are opposite, such as above, one is the negative of the other:

Two vectors are parallel if their directions are either the same or opposite. They are called collinear vectors.

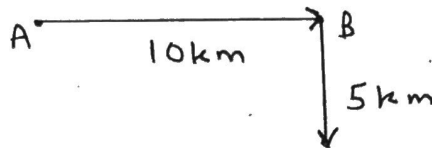
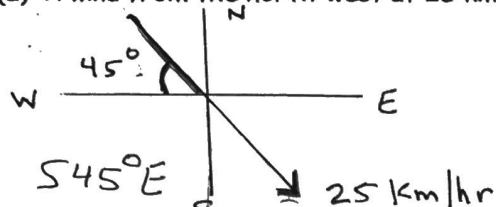
Example #1: State all equal vectors in the diagram below.



$$\begin{cases} \vec{JN} = \vec{KL} \\ \vec{JK} = \vec{NL} \end{cases} \quad \begin{cases} \vec{NJ} = \vec{LK} \\ \vec{KJ} = \vec{LN} \end{cases}$$

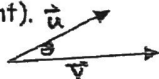
Example #2: Use a suitable scale to draw vectors representing each of the following:

- (a) A wind from the north-west at 25 km/h. (b) A trip of 10 km due east followed by 5 km due south.



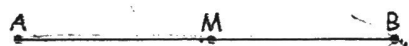
* To specify the direction of a vector, state the angle it makes with another vector or with some given direction such as the horizontal axis, the vertical axis or a compass direction. or bearing.

The angle between two vectors is the angle (no larger than 180 degrees) formed when the vectors are placed tail to tail (ie. Starting at the same point).



Scalar Multiplication:

This is used for 2 vectors that are parallel (where k is a scalar (ie. A real number)).



M is the midpoint of the line segment AB.

Vector equations: $\vec{AM} = \frac{1}{2} \vec{AB}$

$$\vec{AB} = 2\vec{MB}$$

$$\vec{AB} = 2\vec{AM}$$

$$\vec{BM} = -\frac{1}{2} \vec{AB}$$

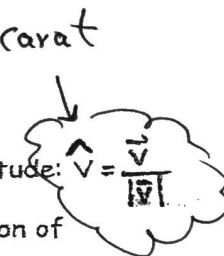
$$\vec{AB} = -2\vec{BM}$$

opposite direction

Unit Vectors:

These are vectors with magnitude one (denoted by a carat (^) placed over the symbol).

1. A unit vector in the direction of any vector can be found by dividing the vector by its magnitude:
2. Any vector can be expressed as the product of its magnitude and a unit vector in the direction of the vector: $\vec{v} = |\vec{v}| \hat{v}$



Zero vector:

The zero vector has a magnitude of 0 (direction is undefined). It is denoted as $\vec{0}$ and is represented by a point.

$$* \quad \vec{u} + \vec{0} = \vec{u}$$

* denoted by

$$* \quad \vec{u} + (-\vec{u}) = \vec{0}$$

a dot

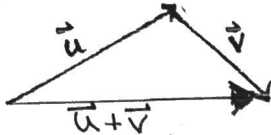


VECTOR LAWS

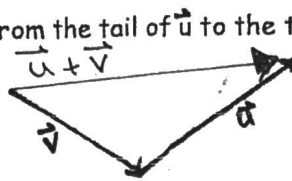
In many applications of vectors, we need to find the combined effect or sum of 2 or more vectors. For example, the effect that wind velocity has on the velocity of an aircraft.

Triangle Law of Vector Addition:

For vectors $\vec{u}$ and $\vec{v}$, the sum (resultant) is a vector from the tail of $\vec{u}$ to the tip of $\vec{v}$, when the tail of $\vec{v}$ is placed at the tip of $\vec{u}$:



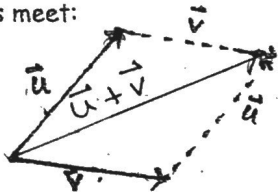
Note: the sum of $\vec{u}$ and $\vec{v}$ is $\vec{u} + \vec{v}$



The order in which we add is not important. Thus, vector addition is commutative: $\vec{u} + \vec{v} = \vec{v} + \vec{u}$

Parallelogram Law of Vector Addition:

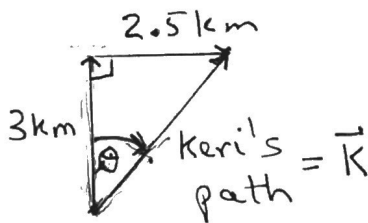
The sum $\vec{u} + \vec{v}$ is the diagonal of the parallelogram from the point where the tails are joined to the point where their heads meet:



* note: The triangle and parallelogram methods are equivalent.

Example #1: Jamie travelled 3 km north then turned to travel 2.5 km due east. Keri leaves after Jamie and decides to take the most direct route to meet Jamie.

(a) Draw a diagram of this situation.



(b) Describe Keri's path in terms of length and direction.

$$|\vec{K}|^2 = 2.5^2 + 3^2 \quad \tan(\theta) = \frac{2.5}{3}$$

$$|\vec{K}| \approx 3.9 \text{ km}$$

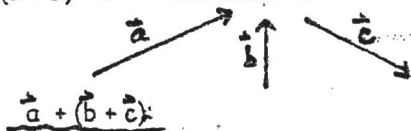
$$\theta \approx 40^\circ$$

$\therefore$ Direction is $N40^\circ E$

OR Bearing 040° .

Associative Law of Addition:

$\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$ This means that we can omit the brackets and write them both equal to $\vec{a} + \vec{b} + \vec{c}$

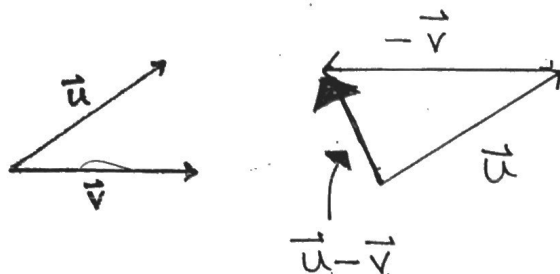


$$(\vec{a} + \vec{b}) + \vec{c}$$

$$\therefore \vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c} = \vec{a} + \vec{b} + \vec{c}$$

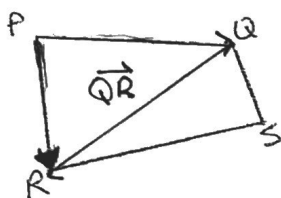
Vector Subtraction:

For vectors $\vec{u}$ and $\vec{v}$, the difference, $\vec{u} - \vec{v}$, is $\vec{u} + (-\vec{v})$. That is, the difference is the same as adding the opposite.



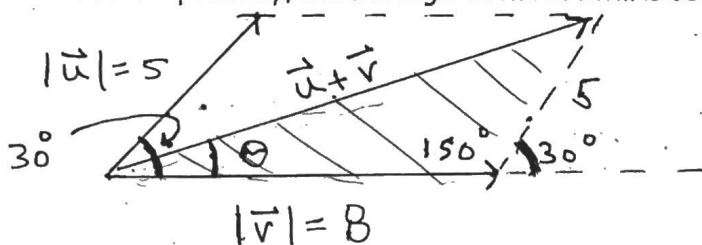
Note: The sum of a vector and its opposite is a vector with zero magnitude. That is, it is a zero vector as $\vec{u} + (-\vec{u}) = \vec{0}$.

Example #2: Given the diagram below, show that $\vec{PQ} - \vec{RQ} = \vec{PR}$.



$$\begin{aligned} \text{L.S.} &= \vec{PQ} - \vec{RQ} \\ &= \vec{PQ} + \vec{QR} \\ &= \vec{PR} \\ &= \text{R.S.} \end{aligned}$$

Example #3: Find the magnitude and direction of the sum of two vectors $\vec{u}$ and $\vec{v}$ if their magnitudes are 5 and 8 units respectively, and the angle between them is 30 degrees.



$$(|\vec{u} + \vec{v}|)^2 = |\vec{u}|^2 + |\vec{v}|^2 - 2|\vec{u}||\vec{v}|\cos(150^\circ)$$

$$= 25 + 64 - 2(5)(8)\cos(150^\circ)$$

$$= 158.28$$

$$\therefore |\vec{u} + \vec{v}| = 12.6 \text{ units}$$

Magnitude

direction

$$\frac{\sin(\theta)}{5} = \frac{\sin(150^\circ)}{12.6}$$

$$\sin(\theta) = \frac{5 \sin 150^\circ}{12.6}$$

$$\therefore \theta = 11.4^\circ \text{ from } \vec{v} \text{ to } \vec{u}.$$